

ONE UNIT OF EXISTENCE

The Life and Death of a Hydrogen Atom
as Told by the Constants That Made It

“What is a hydrogen atom?” is a question that admits an answer at every level of description. This is the answer at the level where the description itself becomes the thing described.

I. Before the Beginning

Before there is a hydrogen atom, there is a lattice. Not a lattice made of anything, but the lattice that makes anything possible.

The lemniscate constant $\varpi \approx 2.622$ is the half-perimeter of the figure-eight curve $r^2 = \cos 2\theta$. It measures the intrinsic geometry of the simplest closed curve that crosses itself: the minimal topology of self-relation. From ϖ , a second constant is born—the bridge constant $G^* = 2\varpi/\sqrt{\pi} \approx 2.959$ —and from these two alone, the ratio

$$\lambda = \frac{\varpi^2}{G^{*2}} = \frac{\pi}{4}$$

generates everything that follows. In the FTD framework, the number π is not treated as fundamental. It is 4λ , four times the ratio of self-intersection to bridging. This is the *ontic ratio*: the single number from which the architecture of discreteness and continuity will be built.

Nothing has happened yet. No particles, no fields, no space. Only the ratio λ and the integers $\{3, 4, 7, 13\}$ that will select, from the continuum of mathematical possibility, one specific universe.

II. The Quark Condensate: Genesis of the Proton

The universe cools. At roughly 10^{-5} seconds after whatever initiated the expansion, the temperature drops below ~ 150 MeV—the QCD confinement scale. Above this energy, quarks move freely in a plasma. Below it, they do not. The transition is a phase change in the strong force, and it is absolute: there will never again, in the entire subsequent history of the cosmos, be a free quark at thermal equilibrium.

Three quarks—two up, one down—are confined into a proton. The confinement is not a cage imposed from outside. It is a topological fact about the $SU(3)$ color field: the flux tubes between quarks carry energy proportional to their length, and stretching them apart costs more energy than creating new quark-antiquark pairs from the vacuum. The system chooses the lowest-energy configuration, which is a bound state.

The number of colors is not arbitrary. In FTD, the master quadratic

$$x^2 - 16 G^{*2} x + 16 G^{*3} = 0$$

has two roots. The larger root gives $1/\alpha$. The smaller root, $x_- \approx 3.024$, has integer part 3: this is N_c , the number of color charges. The one-loop QCD beta-function coefficient $b_3 = (11N_c - 2N_f)/3 = 7$ (for $N_f = 6$ quark flavors) then sets the running of the strong coupling. The proton's confinement is, in this reading, controlled by the same quadratic that will later determine its electromagnetic binding to an electron.

The proton's mass is approximately 938 MeV. Almost none of this comes from the quarks themselves (the up and down quarks have masses of a few MeV each). The overwhelming majority—over 99%—is the kinetic and binding energy of the strong field, whose confinement scale Λ_{QCD} is set by α_s running under $b_3 = 7$. The proton is, in this precise sense, made of motion, not of matter.

This is relevant. The proton is already a discrete-to-continuous bridge: a discrete, countable object (baryon number = 1) whose internal structure is a continuous quantum field theory. The proton will outlive every star. Its lifetime exceeds 10^{34} years [3]. For the purposes of this story, it is eternal.

III. Recombination: The Electron Finds Its Home

For the first 380,000 years, the universe is a plasma. Protons exist, electrons exist, photons exist, but the temperature is too high for neutral atoms to form. Every time an electron settles into orbit around a proton, a photon with energy above 13.6 eV strikes it free. The universe is opaque: photons scatter constantly off the dense sea of charged particles.

Then the temperature drops below ~ 3000 K, and the photon bath no longer carries enough energy to ionize hydrogen. In a cosmic eyeblink, nearly every proton captures an electron. The universe becomes transparent.

This is the moment our hydrogen atom is born.

The electron does not orbit the proton like a planet. It occupies a standing wave, a stationary state of the Schrödinger equation:

$$E_n = -\frac{\alpha^2 m_e c^2}{2n^2}, \quad n = 1, 2, 3, \dots$$

where α is the fine-structure constant.

Here is where the story turns. The fine-structure constant is the number that determines the strength of the electromagnetic interaction—the force that holds our atom together. In the FTD framework, it emerges from the master quadratic whose coefficients are built entirely from G^* (and therefore from ϖ), yielding

$$\frac{1}{\alpha} \approx 137.036,$$

at a precision of 0.21 parts per trillion [1]. The four integers $\{3, 4, 7, 13\}$ select this quadratic from the space of all possible quadratics over the lemniscatic lattice. The hydrogen atom's entire energy spectrum—every line in the Balmer series, every transition that Bohr, Sommerfeld, and Dirac painstakingly catalogued—follows from this single algebraic object.

As a concrete checkpoint: the ground-state binding energy is

$$|E_1| = \frac{\alpha^2 m_e c^2}{2} = \frac{(1/137.036)^2 \times 0.51100 \text{ MeV}}{2} = 13.606 \text{ eV},$$

matching the measured ionization energy of hydrogen to five significant figures. This is not a fit— α and m_e are both determined within FTD; the 13.6 eV follows as output.

The electron settles into the $n = 1$ ground state. Its “orbital” is a spherically symmetric cloud, densest at the nucleus, decaying exponentially with a characteristic scale set by the Bohr radius:

$$a_0 = \frac{\hbar}{\alpha m_e c} = \frac{1}{\alpha} \cdot \frac{\hbar}{m_e c} \approx 0.529 \text{ \AA}.$$

Notice: a_0 is the Compton wavelength of the electron divided by α . The Bohr radius is not an independent parameter. It is α ’s signature in configuration space, and α is G^* ’s signature in coupling space.

The atom now exists. It is one proton, one electron, and a quantum field that binds them. It weighs slightly less than its parts: the mass deficit is the binding energy, $13.6 \text{ eV}/c^2$, a tiny fraction of the total 938 MeV . It is electrically neutral. It is the simplest bound state in nature.

IV. The Discrete Spectrum: A Sum over Integers

Our hydrogen atom sits in its ground state. It is silent, invisible, inert. Then a photon arrives.

If the photon carries exactly the right energy— $E_2 - E_1 = 10.2 \text{ eV}$ —the atom absorbs it and the electron jumps to $n = 2$. Not approximately. Exactly. The atom either absorbs the photon or it does not. There is no half-transition. Discreteness is absolute.

The photon energies that the atom can absorb form a discrete set:

$$E_{n \rightarrow m} = \frac{\alpha^2 m_e c^2}{2} \left(\frac{1}{n^2} - \frac{1}{m^2} \right), \quad n < m.$$

This is a sum over integers. The same *structural pattern*—discrete indices bridged to a continuous closed form by an analytic continuation—appears in the identity

$$\sum_{n=0}^{\infty} \frac{1}{(n^2 + 1)^2} = \frac{2 + \pi \coth \pi + \pi^2 \operatorname{csch}^2 \pi}{4} = \frac{2 + 4\lambda \coth(4\lambda) + 16\lambda^2 \operatorname{csch}^2(4\lambda)}{4}.$$

The analogy is one of *form*, not of mechanism: the Rydberg series arises from the Coulomb Green’s function, while the sum above passes through the Gaussian lattice $\mathbb{Z}[i]$. But the architectural motif is the same. Discrete energy levels are the integers of the atom’s inner life. The electromagnetic field in which they are embedded is the continuous medium. The coupling constant α —determined by G^* —is the bridge between them.

The hydrogen emission spectrum, recorded on a photographic plate, is a set of sharp lines: discrete, countable, unmistakable. But the underlying field theory that produces those lines (quantum electrodynamics) is continuous, built on integrals over all of momentum space. The sharp lines emerge from the continuum exactly as the closed-form answer emerges from the infinite discrete sum—different mechanisms, same structural crossing from $\sum$ to $\int$.

In both cases, the bridge is the same constant in different clothing.

V. The Lamb Shift: Where the Vacuum Speaks

If the hydrogen atom were governed by the Dirac equation alone, the $2S_{1/2}$ and $2P_{1/2}$ states would have exactly the same energy. They do not. The $2S_{1/2}$ state is shifted upward by about 1058 MHz —the Lamb shift, measured by Willis Lamb in 1947 [2].

The shift comes from the vacuum. Even in its ground state, the electromagnetic field fluctuates: virtual photons are constantly created and annihilated, and the electron interacts with them. These vacuum fluctuations slightly smear the electron’s position, effectively making it “see” a slightly different nuclear potential than it would classically. The S states, which have nonzero probability density at the nucleus, are more affected than P states, which vanish there.

The leading-order Lamb shift scales as α^5 : five powers of the fine-structure constant, five powers of G^* ’s imprint. Each power of α represents one more layer of interaction between the discrete (the atom’s energy levels) and the continuous (the quantum vacuum). The Lamb shift is the discrete-to-continuous bridge made measurable: the atom’s sharp energy levels responding to the hum of the infinite continuum beneath them.

This was the measurement that confirmed quantum electrodynamics. The vacuum is not empty. The vacuum is the continuum from which discreteness is projected.

VI. The Hyperfine Transition: The Quietest Sound

The proton has spin. The electron has spin. When aligned, the total energy is slightly higher than when anti-aligned. The difference is the hyperfine splitting: $5.9 \mu\text{eV}$, corresponding to a photon frequency of 1420.405 MHz, or a wavelength of 21 cm.

This is the most famous frequency in radio astronomy. Neutral hydrogen throughout the universe occasionally undergoes the spin-flip transition, emitting a 21-cm photon. The transition is fantastically slow: the spontaneous lifetime is about 10^7 years. A single hydrogen atom in its excited hyperfine state will, on average, wait ten million years before emitting.

But there are $\sim 10^{80}$ hydrogen atoms in the observable universe. So the 21-cm line is everywhere, all the time, in every direction. It is the sound of the universe breathing.

The hyperfine splitting scales as $\alpha^4 (m_e/m_p)$: the fine-structure constant to the fourth power, modulated by the electron-to-proton mass ratio. The mass ratio introduces the strong force (since m_p is set by QCD, whose confinement scale runs under $b_3 = 7$), so the hyperfine line is a meeting point of two of the fundamental interactions, both encoded, in the FTD picture, in the algebraic structure of the lemniscatic lattice.

VII. Stellar Fusion: Death by Continuity

Our hydrogen atom drifts for hundreds of millions of years, part of a diffuse cloud in intergalactic space. Then gravity gathers the cloud. It contracts, heats, and eventually the core temperature reaches $\sim 10^7$ K.

At this temperature, protons have enough kinetic energy to overcome their mutual Coulomb repulsion and approach within ~ 1 fm of each other, where the strong nuclear force can act. Two protons fuse: one undergoes beta-plus decay (converting to a neutron via the weak force), and the result is a deuteron, a positron, and a neutrino.

Our hydrogen atom’s proton has just become part of a deuterium nucleus. The atom, as a distinct entity, is gone.

The probability of this fusion event is controlled by quantum tunneling through the Coulomb barrier. The tunneling rate depends exponentially on the Gamow energy:

$$E_G = 2 m_r c^2 (\pi \alpha Z_1 Z_2)^2,$$

where α appears again. The exponential sensitivity means that even a small change in α would either extinguish all stars or cause them to burn through their fuel in moments. The

value of α implied by the FTD master quadratic happens to fall in the narrow window compatible with stellar lifetimes of $\sim 10^{10}$ years, long enough for complexity to emerge on their planets.

This is not a teleological claim. It is an observational constraint. Whether the compatibility of FTD's α with stellar longevity reflects selection, coincidence, or necessity is an open question. FTD provides the value; it does not (yet) explain why the value is life-permitting.

VIII. The Fusion Chain: From Hydrogen to Iron

The deuteron that was once our atom's proton captures another proton to form helium-3, and two helium-3 nuclei fuse to form helium-4 plus two protons returned to the plasma. This is the pp chain, the dominant energy source in solar-type stars.

The mass of helium-4 is less than the mass of four protons by 26.7 MeV. This mass deficit, converted to energy via $E = mc^2$, is what powers the star. The sun converts about 600 million tonnes of hydrogen into helium every second. Our atom's proton contributed its share: 6.7 MeV per nucleon, released as photons and neutrinos.

In more massive stars, the chain continues: helium to carbon (via the triple-alpha process), carbon to oxygen, oxygen to neon, neon to silicon, silicon to iron. At iron, fusion becomes endothermic. The chain stops. The star has converted the maximum possible fraction of its rest mass into binding energy, and the nucleus of iron-56 sits at the bottom of the valley of stability.

Each step in this chain is a discrete nuclear transition, but the stellar interior in which it occurs is a continuous plasma. The rates of these transitions are set by coupling constants (strong, weak, electromagnetic) that are, in the FTD picture, determined by the integers $\{3, 4, 7, 13\}$ acting on the lemniscatic lattice. The entire periodic table is a consequence of these four numbers.

IX. Supernova: The Return

If the star is massive enough, the iron core collapses. In less than a second, a volume the size of the Earth is compressed to a sphere 20 km across. The collapse is halted by neutron degeneracy pressure—a quantum effect, the Pauli exclusion principle forbidding two neutrons from occupying the same state. The infalling matter bounces off this incompressible core and an outward shock wave tears the star apart.

This is a supernova. It releases $\sim 3 \times 10^{46}$ J in a few seconds, briefly outshining its entire host galaxy. The shock wave disperses the star's outer layers—including all the elements heavier than helium that were forged in the core—back into the interstellar medium.

Among the debris: new hydrogen atoms, formed as the expanding ejecta cool and free protons capture electrons from the thinning plasma. The cycle is ready to begin again.

But our particular proton is likely buried deep, fused into iron or heavier elements synthesized in the explosion's neutron flux. It is no longer hydrogen. Its identity as a distinct baryon persists (baryon number is conserved, as far as any experiment has ever shown), but its context has changed irreversibly.

X. Annihilation: The Final Bridge

There is one more chapter, though it may never be written.

If baryon number is not exactly conserved—if grand unified theories are correct and the proton eventually decays—then the last trace of our hydrogen atom will vanish. Many GUT models predict the dominant channel $p \rightarrow e^+ + \pi^0$, followed by $\pi^0 \rightarrow 2\gamma$ and e^+ annihilating with an ambient electron to produce two more photons. The entire rest mass of the proton is converted to radiation. Whether this channel is realized, and whether FTD’s algebraic structure permits or forbids it, remains an open question.

The proton’s lifetime is at least 10^{34} years [3], far longer than the current age of the universe (1.4×10^{10} years). If it decays at all, it will do so in an era when every star has burned out, every black hole has evaporated, and the universe consists of nothing but a thin, cooling gas of photons, neutrinos, and the occasional lonely electron or positron.

At that moment, the last discrete object—the proton, baryon number one—crosses the final bridge into the continuum of radiation. Photons have no rest mass, no charge, no baryon number. They are pure field, pure wave, pure continuity. The integer becomes the irrational. The lattice point dissolves into the plane.

$$\Sigma \rightarrow \int$$

Coda: The Ratio Remains

Every stage of this story was governed by the same architecture:

Event	Discrete	Continuous
Quark confinement	baryon number	QCD flux tubes
Recombination	electron capture	photon field
Absorption/emission	energy levels E_n	electromagnetic field
Lamb shift	atomic state	vacuum fluctuations
Hyperfine transition	spin states	radio-frequency field
Stellar fusion	nuclear binding	thermal plasma
Proton decay	baryon number $\rightarrow 0$	radiation field

At every transition, the bridge between the discrete side and the continuous side is controlled by coupling constants that are, in the FTD framework, determined by ϖ , G^* , and the integers $\{3, 4, 7, 13\}$.

The hydrogen atom does not “know” about the lemniscatic lattice. It does not need to. It simply *is* what the lattice describes, projected into three spatial dimensions and one temporal one, bound by a coupling constant that is a root of a quadratic over G^* , radiating at frequencies that are integer multiples of $\alpha^2 m_e c^2$, and destined to dissolve back into the continuum from which it was projected.

The ratio $\lambda = \varpi^2/G^{*2}$ was there before the atom, is there during the atom, and will be there after the atom. The atom is a transient excitation. The constants are not made of atoms.

The bridge is not π . The bridge is G^* .

And the story of one hydrogen atom is the story of every hydrogen atom, which is the story of every discrete thing that ever existed in a continuous universe.

Foundational Ternary Dynamics: a framework, not a faith.

References

- [1] E. Tiesinga *et al.*, “CODATA recommended values of the fundamental physical constants: 2022,” *Rev. Mod. Phys.* **96** (2024), 015002.
- [2] W. E. Lamb and R. C. Retherford, “Fine structure of the hydrogen atom by a microwave method,” *Phys. Rev.* **72** (1947), 241–243.
- [3] Super-Kamiokande Collaboration, “Search for proton decay via $p \rightarrow e^+\pi^0$ and $p \rightarrow \mu^+\pi^0$,” *Phys. Rev. D* **95** (2017), 012004.